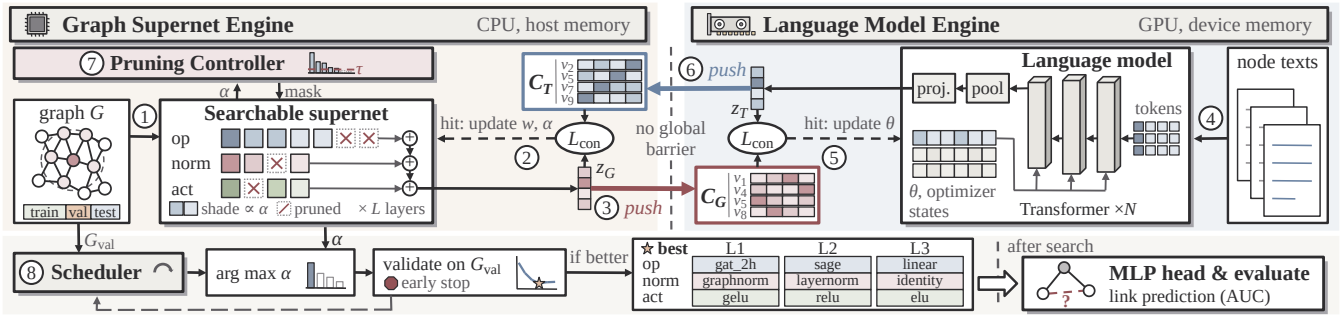

Graph Supernet Engine
CPU, host memory
Language Model Engine
GPU, device memory
⑦ Pruning Controller
τ
α
mask
graph G
train val test
Searchable supernet
op
norm
act
shade $\propto \alpha$
pruned
$\times L$ layers
hit: update w, α
L_{con}
C_T
v_2 v_5 v_7 v_9
z_G
③ push
⑥ push
z_T
no global barrier
hit: update θ
C_G
v_1 v_4 v_5 v_8
Language model
proj.
pool
tokens
θ, optimizer states
Transformer $\times N$
node texts
G_{val}
⑧ Scheduler
arg max α
validate on G_{val}
early stop
if better
☆ best
L1
L2
L3
op
gat_2h
sage
linear
norm
graphnorm
layernorm
identity
act
gelu
relu
elu
after search
MLP head & evaluate
link prediction (AUC)